

Restructured Introduction — Outline

Geolocation for Lapse Prediction in AUD (Psychological Medicine AI call)

2026-09-17

¶1 — AUD as a chronic, relapsing disorder; lapses as a tractable target

Topic sentence: *Alcohol use disorder is common and, for many, follows a chronic course in which hard-won recovery is repeatedly interrupted by returns to use.*

- AUD prevalence (1 in 10 U.S. adults, 2022).
- Chronic, relapsing course; recovery interspersed with returns to harmful use.
- Lapses (single goal-inconsistent use) temporally precede relapse; clearly defined and observable → suitable early-intervention target.
- Precipitants vary between and within people and are hard to anticipate → motivates ongoing, individualized monitoring.

¶2 — The clinical problem: continuous risk monitoring can't scale through clinicians alone

Topic sentence: *Sustained support for a lifelong, fluctuating risk process exceeds what clinicians can feasibly provide, creating a clear role for algorithmic decision support.*

- Ideal ongoing support = a continuous risk monitoring and support system.
- Real-time, day-by-day risk detection is infeasible in an overburdened addiction treatment system.
- Frames the clinical decision-making need the call emphasizes: scalable, sustainable risk stratification to inform when/whom to support.
- Two requirements previewed: temporally precise risk-relevant data + a scalable modeling engine.

¶3 — Personal sensing + machine learning as the enabling technology

Topic sentence: *Personal sensing paired with machine learning offers a route to risk monitoring that is both ecologically valid and scalable.*

- Personal sensing = passively/actively collected data from ubiquitous devices (smartphones, wearables), including EMA.
- Unobtrusive, continuous, low behavior change → sustainable and ecologically valid.

- ML uncovers relationships between antecedent signals and lapse; enables scalability beyond clinician capacity.

¶4 — Prior EMA work and its limitations

Topic sentence: *Brief daily surveys have established that self-reported states can forecast lapse, but self-report carries feasibility and coverage limits that constrain a monitoring system.*

- EMA predicts future alcohol use (Wyant et al., 2023); captures self-efficacy, past use, stress, mood.
- Limits: requires insight; burden for people with jobs/comorbidities; fewer/inaccurate responses degrade prediction.
- Discrete sampling vs. a dynamic recovery process → information loss between surveys.
- Motivates a data source with finer temporal granularity and lower burden.

Subsection: *Geolocation Data for Risk Monitoring*

¶5 — Geolocation's advantages over EMA

Topic sentence: *Geolocation data can be sampled continuously with little user input, addressing several of EMA's core limitations.*

- Latitude/longitude sampled at regular intervals → greater temporal sensitivity/specificity than EMA.
- Passive after setup; often collected by default → low burden.
- Potential equity benefit: accessible to people who can't complete multiple daily surveys (sets up fairness theme).

¶6 — Why location is substantively linked to lapse

Topic sentence: *Beyond convenience, where a person is carries genuine signal about lapse risk.*

- Environmental cues and perceived riskiness of settings shape lapse (Janak; Walton).
- Location-targeted coping is already embedded in treatments (e.g., mindfulness-based relapse prevention).
- Geolocation flagged as valuable for understanding and treating substance use (Stahler).

¶7 — Raw geolocation features from the digital phenotyping literature

Topic sentence: *One body of work derives "raw" movement features that quantify behavioral patterns linked to mental health.*

- Raw features = derived from basic movement (location variance, entropy, time at home, transit, circadian rhythmicity).

- Linked to loneliness/isolation, depression, negative symptoms of schizophrenia, mood; used to predict symptom emergence.
- Strength: minimal added input; limitation: strip out environmental context (two identical trajectories can mean very different exposures).

¶8 — Contextualized features that restore person-specific meaning

Topic sentence: *A complementary approach enriches location with brief context about what specific places mean to a person.*

- Context-based features add environmental qualities (SES, crime) — but broad-strokes labels can misclassify person-specific risk (Mennis & Mason “Ecological Interview” example).
- Contextualized features = geolocation + brief intermittent surveys about frequented locations (bridges EMA’s personalization with geolocation’s low burden).
- Precedent: geographical momentary assessment; recovery apps using self-identified risky locations / geofences.
- (*Intervenability/actionable-recommendation payoff deferred to Discussion, per current draft.*)

¶9 — Responsible AI as a requirement, not an afterthought

Topic sentence: *For a prediction model to be clinically meaningful, it must be evaluated not only for accuracy but for equity and interpretability across the people it is meant to serve.*

- Risk models can perform unevenly across subgroups defined by race, age, income, education, sex → risk of amplifying disparities in care.
- Passive geolocation may improve access for under-resourced groups but could also encode mobility patterns that differ systematically by demographics → fairness must be tested, not assumed.
- Explainability (which features drive risk) is prerequisite for clinician trust and for eventual actionable use.
- Positions the study within the call’s responsible-AI scope (fairness, bias, explainability, reporting standards).

¶10 — The gap

Topic sentence: *Despite this promise, no work has tested the predictive value of geolocation for alcohol lapse, nor whether such prediction is equitable and interpretable.*

- Prior geolocation-mental-health work is largely descriptive/associational, not predictive of AUD lapse.
- Combining raw + contextualized features could capture heterogeneous precipitants (social isolation, alcohol presence, affect tied to place).
- Open questions: Does geolocation predict lapse? Does it add value beyond demographics? Is performance fair across subgroups? Which features carry signal?

Subsection: *The Current Study*

¶11 — Aims and contributions

Topic sentence: *This study uses geolocation data and machine learning to predict next-day lapse in adults with AUD pursuing abstinence, addressing several gaps aligned with clinically meaningful, responsible AI.*

- Novel use of geolocation to predict lapse, extending the lab’s EMA work.
- Raw + contextualized features to cast a wide net over precipitants.
- Evaluate incremental value over demographic baseline.
- **Assess algorithmic fairness across subgroups** (elevated to a stated aim).
- Quantify feature importance (explanatory goal supporting eventual clinical use).
- Frame as an initial step toward a continuous risk monitoring and support system.